

Nilesh Chidambaram

Software Developer · IT Graduate · Mumbai, India

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PROFILE

B.Sc. IT graduate with hands-on experience in Python, AI/ML, Generative AI, and full-stack web development. Built ML-based anomaly detection systems and AI-powered applications using LLMs and REST APIs, along with web applications using Next.js and TypeScript. Experienced in data preparation, model experimentation, testing, and documentation, with a focus on practical end-to-end solutions.

EXPERIENCE

AI Variant

Nov 2025

AI/ML Intern

Mumbai, India

- Collaborated on a Real-Time Intrusion Detection System, contributing to data preparation, model experimentation, and testing within a 2-person Agile team.
- Applied anomaly-detection techniques and ML workflows, using AI-assisted development tools to accelerate iteration.

PROJECTS

Note-Sync — Real-Time Collaborative Editor | [Github](#)

TypeScript · React · Next.js · Tailwind CSS · TipTap · Liveblocks · Convex · Clerk · Gemini AI · Tesseract.js

- Implemented a full-stack collaborative rich-text editor with live cursors and threaded comments using Liveblocks WebSocket architecture, and extended TipTap with 2 custom extensions (font-size and line-height controls) via custom ProseMirror attribute schemas and chained commands.
- Integrated Google Gemini AI via REST API to power 5 AI-assisted writing actions (improve writing, rewrite, summarize, shorten, lengthen) alongside a context-aware LLM chat assistant.
- Added in-browser OCR (Tesseract.js) supporting both PDF and image inputs for document text extraction, secured with Clerk authentication and organization-level access control, backed by Convex as the reactive database.

JobLens — Job Application Tracker & Resume Matcher | [Github](#)

Python · OOP · JSON · PyPDF · Matplotlib · Pytest

- Developed a Python CLI application for end-to-end job-application tracking across 5 stages (Applied, Online Assessment, Interview, Offer, Rejected), with CRUD operations, search, and persistent JSON storage.
- Engineered a PDF-based resume-matching engine (PyPDF, OOP) that extracts skills from resumes and job descriptions to generate a normalized 0–100% match score and flag missing skills.
- Validated core logic with an 18-test Pytest suite covering input validation, JSON persistence, file-error handling, and skill-matching calculations, and visualized pipeline status through a Matplotlib analytics dashboard.

AI-Powered Network Intrusion Detection System — Team Project | [Github](#)

Python · Scapy · River · Gradio · Plotly · FPDF2

- Built a real-time Network Intrusion Detection System with a 2-person team, capturing live network traffic via Scapy and detecting anomalies using River's Half Space Trees for online, incremental learning.
- Designed an interactive Gradio dashboard with real-time Plotly visualizations, protocol distribution analysis, protocol/IP-based packet filtering, a live detection log, and PDF incident-report export via FPDF2.

EDUCATION

Guru Nanak Khalsa College of Science, Arts & Commerce

2023 – 2026

B.Sc. Information Technology · Mumbai, India

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript
- **Web:** React.js, Next.js, Tailwind CSS, TipTap, HTML, CSS, Responsive Design
- **AI/ML:** ML Workflows, Data Preprocessing, Anomaly Detection, Generative AI / LLM Integration (Gemini AI)
- **Backend/DB:** REST APIs, MySQL, MongoDB, Convex, JSON
- **Python Libraries:** OOP, PyPDF, Matplotlib, Pytest, Pandas
- **Tools:** Git, GitHub, Liveblocks, Clerk, VS Code, Linux (Ubuntu), Tesseract.js, Tableau

CERTIFICATIONS

- HarvardX — CS50's Introduction to Programming with Python
- IBM SkillsBuild Winter Certification 2025 — Data Analytics & Business Intelligence Lab
- Accenture Developer and Technology Job Simulation — Forage
- Artificial Intelligence & Machine Learning — ExcelR Edtech Pvt. Ltd. (2025)